AP PRECALCULUS

Unit 1 • Lesson 1.6 notes

Lesson 1.6 • Polynomial Functions and End Behavior

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Describe the end behavior of a polynomial with limit notation.
- Determine end behavior from the degree and the sign of the leading coefficient.

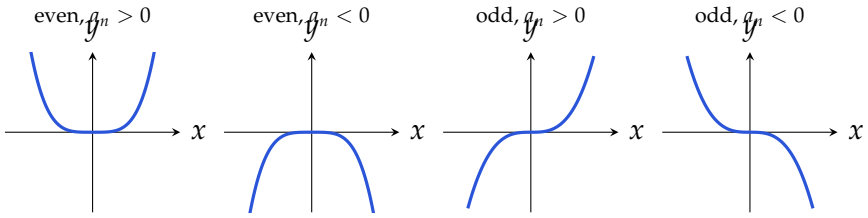
Vocabulary. end behavior, leading term, limit at infinity, $\lim_{x\to\infty}$, $\lim_{x\to-\infty}$.

1. Limit notation

$\lim_{x\to\infty} p(x) = \infty$ means: as x increases without bound, $p(x)$ increases without bound. Similar statements describe $x \to -\infty$ and $-\infty$ outputs.

2. The leading term decides

For large $|x|$, the leading term $a_n x^n$ is much larger than all other terms combined, so p has the same end behavior as $a_n x^n$.



Example 1

Describe the end behavior of $p(x) = 3x^2 - 2x^5 + 7$.

Solution. Leading term $-2x^5$ (the highest power, not the first written term): odd degree, negative coefficient. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$.

Example 2 • Factored form

Describe the end behavior of $q(x) = -x(x-2)^2(3-x)$.

Solution. Multiply the leading terms of the factors: $-x \cdot x^2 \cdot (-x) = x^4$. Even degree, positive: $\lim_{x\to\pm\infty} q(x) = \infty$.

Common mistakes

- Taking the first term written as the leading term.
- In factored form, forgetting the sign hidden in a factor such as $(3-x)$, whose leading term is $-x$.

Quick check

1. $\lim_{x\to-\infty}(4-x^3) =$? 2. Which degree and sign give $\lim_{x\to\pm\infty} p(x) = -\infty$?

Answers. 1. ∞. 2. Even degree, negative leading coefficient.